Day 7 – Array Practice (10 Problems)

Problem 1 – Reverse Array

Given an array, usko reverse karo.

Example

Input: [1,2,3,4,5]
Output: [5,4,3,2,1]

Problem 2 – Find Maximum Element

Array me **sabse bada element** find karo.

Example

Input: [5,9,2,7]
Output: 9

Problem 3 – Find Minimum Element

Array me **sabse chhota element** find karo.

Example

Input: [5,9,2,7]
Output: 2

Problem 4 – Linear Search

Given element array me exist karta hai ya nahi check karo.

Example

Array = [10,20,30,40]
Key = 30
Output = Found at index 2

Problem 5 – Binary Search

Sorted array me element search karo.

Example

Array = [2,4,6,8,10]
Key = 8
Output = Index 3

Problem 6 – Insert Element in Array

Array ke **given position par element insert** karo.

Example

Array = [10,20,30,40]
Insert = 25 at index 2

Output:
[10,20,25,30,40]

Problem 7 – Delete Element from Array

Array se **given position ka element delete** karo.

Example

Array = [10,20,30,40]
Delete index = 1

Output:
[10,30,40]

Problem 8 – Two Sum Problem

Array me do numbers find karo jinka sum **target ke equal ho**.

Example

Array = [2,7,11,15]
Target = 9

Output:
Indices: 0,1

Problem 9 – Majority Element

Array me wo element find karo jo **n/2 se zyada baar appear ho**.

Example

Array = [2,2,1,1,2,2,3]

Output:
2

Problem 10 – Subarray Sum

Array ke **sabhi subarray ka sum print karo**.

Example

Array = [1,2,3]

Output

1
3
6
2
5
3

✓ Coding of Each Question

Problem 1 – Reverse Array

```
public class ReverseArray {
    public static void main(String[] args) {

        int[] arr = {1,2,3,4,5};

        int start = 0;
        int end = arr.length - 1;

        while(start < end){
            int temp = arr[start];
            arr[start] = arr[end];
            arr[end] = temp;

            start++;
            end--;
        }

        for(int num : arr){
            System.out.print(num + " ");
        }
    }
}
```

Problem 2 – Find Maximum Element

```
public class MaxElement {
    public static void main(String[] args) {

        int[] arr = {5,9,2,7};

        int max = arr[0];

        for(int i = 1; i < arr.length; i++){
            if(arr[i] > max){
                max = arr[i];
            }
        }

        System.out.println("Maximum = " + max);
    }
}
```

Problem 3 – Find Minimum Element

```
public class MinElement {
    public static void main(String[] args) {

        int[] arr = {5,9,2,7};

        int min = arr[0];

        for(int i = 1; i < arr.length; i++){
            if(arr[i] < min){
                min = arr[i];
            }
        }

        System.out.println("Minimum = " + min);
    }
}
```

Problem 4 – Linear Search

```
public class LinearSearch {
    public static void main(String[] args) {

        int[] arr = {10,20,30,40};
        int key = 30;

        for(int i = 0; i < arr.length; i++){
            if(arr[i] == key){
                System.out.println("Found at index " + i);
                return;
            }
        }

        System.out.println("Not Found");
    }
}
```

Problem 5 – Binary Search

```
public class BinarySearch {
   public static void main(String[] args) {

       int[] arr = {2,4,6,8,10};
       int key = 8;

       int left = 0;
       int right = arr.length - 1;

       while(left <= right){

           int mid = (left + right) / 2;

           if(arr[mid] == key){
               System.out.println("Element found at index " + mid);
               return;
           }

           else if(arr[mid] < key){
               left = mid + 1;
           }

           else{
               right = mid - 1;
           }
       }

       System.out.println("Element not found");
   }
}
```

Problem 6 – Insert Element in Array

```
public class InsertElement {
   public static void main(String[] args) {

       int[] arr = {10,20,30,40,0};
       int size = 4;
       int position = 2;
       int element = 25;

       for(int i = size; i > position; i--){
           arr[i] = arr[i-1];
       }

       arr[position] = element;
       size++;

       for(int i = 0; i < size; i++){
           System.out.print(arr[i] + " ");
       }
   }
}
```

Problem 7 – Delete Element from Array

```
public class DeleteElement {
    public static void main(String[] args) {

        int[] arr = {10,20,30,40};
        int size = 4;
        int position = 1;

        for(int i = position; i < size - 1; i++){
            arr[i] = arr[i+1];
        }

        size--;

        for(int i = 0; i < size; i++){
            System.out.print(arr[i] + " ");
        }
    }
}
```

Problem 8 – Two Sum (Brute Force)

```
public class TwoSum {
    public static void main(String[] args) {

        int[] arr = {2,7,11,15};
        int target = 9;

        for(int i = 0; i < arr.length; i++){
            for(int j = i+1; j < arr.length; j++){

                if(arr[i] + arr[j] == target){
                    System.out.println("Indices: " + i + " , " + j);
                }

            }
        }
    }
}
```

Problem 9 – Majority Element

```
1. public class MajorityElement {
    public static void main(String[] args) {

        int[] arr = {2,2,1,1,2,2,3};
        int n = arr.length;

        for(int i = 0; i < n; i++){
            int count = 0;

            for(int j = 0; j < n; j++){
                if(arr[i] == arr[j]){
                    count++;
                }
            }

            if(count > n/2){
                System.out.println("Majority Element = " + arr[i]);
                break;
            }
        }
    }
}
```

Problem 10 – Subarray Sum

```
public class SubarraySum {
    public static void main(String[] args) {

        int[] arr = {1,2,3};

        for(int i = 0; i < arr.length; i++){

            int sum = 0;

            for(int j = i; j < arr.length; j++){

                sum = sum + arr[j];

                System.out.println("Subarray sum = " + sum);

            }

        }
    }
}
```